Properties of inequality

1 Write the inequality $18 \leq 23$ in words.

2 For each of the following scenarios:

i Write the new inequality produced.

ii Does the new inequality still hold true?

a Adding 6 to both sides of the inequality $7 < 10$.

b Adding 3 to both sides of the inequality $13 > 10$.

c Adding 2 to both sides of the inequality $-10 < -7$.

d Adding 3 to both sides of the inequality $4 > -2$.

e Subtracting 5 from both sides of the inequality $1 < 3$.

f Adding -3 to both sides of the inequality $12 > 10$.

g Multiplying both sides of the inequality $5 < 7$ by 2.

h Multiply both sides of the inequality $6 < 10$ by -4 but not changing the sign of the inequality.

i Dividing both sides of the inequality $5 > 2$ by 6.

j Dividing both sides of the inequality $8 < 12$ by -4 but not changing the sign of the inequality.

3 If $a < b$ is true, explain why the following three statements are true:

- $a + 1 < b + 1$
- $a - 1 < b - 1$
- $a + n < b + n$ for any integer n

4 Given a is an integer less than zero, and $x \geq 9$, determine whether the following inequalities are true or false statements.

a $\dfrac{x}{a} \geq \dfrac{9}{a}$

b $x \times a \geq 9 \times a$

c $x - a \geq 9 - a$

d $-x \times a \geq -9 \times a$

5 Given a is an integer greater than zero, and $x \leq 3$, determine whether the following inequalities are true or false statements.

a $\dfrac{x}{a} \leq \dfrac{3}{a}$

b $x \times a \leq 3 \times a$

c $-x \times a \leq -3 \times a$

d $x - a \leq 3 - a$

6 Given a is a positive integer, determine whether the following operations maintain the inequality $x \geq 6$.

a Adding or subtracting a to both sides of the inequality.

b Multiplying or dividing both sides of the inequality by $-a$.

c Multiplying or dividing both sides of the inequality by a.

d Adding or subtracting $-a$ to both sides of the inequality.

7 Given a is a negative integer, determine whether the following operations maintain the inequality $x \leq 2$.

a Adding or subtracting a to both sides of the inequality.

b Multiplying or dividing both sides of the inequality by $-a$.

c Multiplying or dividing both sides of the inequality by a.

d Adding or subtracting $-a$ to both sides of the inequality.

8 Consider the lines of working below.

$$\begin{aligned} a &< b \\ a - b &< b - b \\ a - b &< 0 \\ a - b - a &< 0 - a \\ -b &< -a \end{aligned}$$

Determine whether following statements are true or false.

a Multiplying an inequality by -1 does not affect the way the inequality is read.

b Multiplying an inequality by -1 reverses the direction of the inequality sign.

c If $a < b$, then $-a > -b$.

d If $a < b$, then $-a < -b$.

9 Given $a < b$, show that $3a < 3b$ using addition.